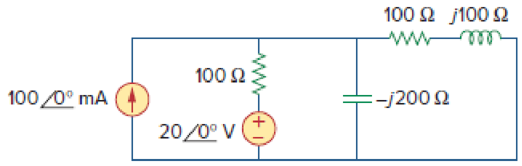
Chapter 11, Problem 59P

Problem

Calculate the reactive power in the inductor and capacitor in the circuit of Fig. 11.78.

Figure 11.78:



Step-by-step solution

Step 1 of 9

Refer to Figure 11.78 in the text book.

Consider the circuit diagram shown in Figure 1.

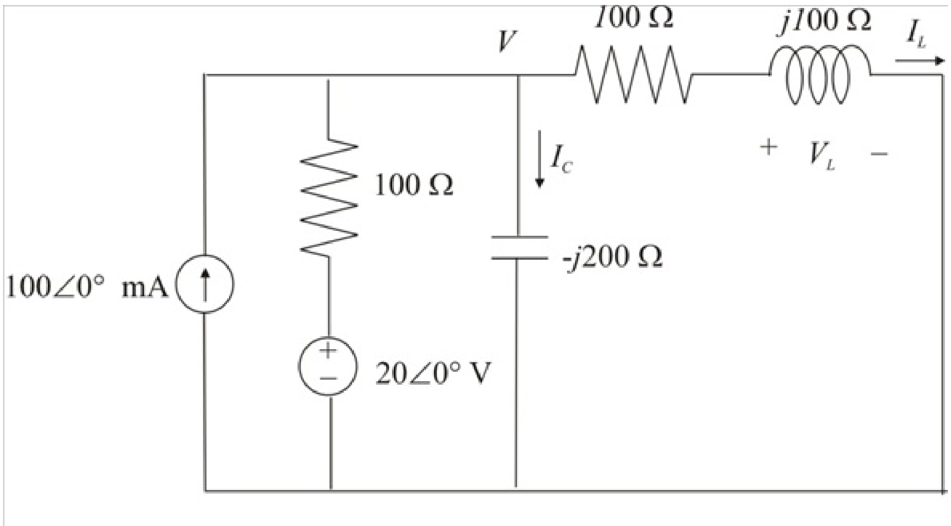


Figure 1

Step 2 of 9

Hand calculations:

Apply Kirchhoff's Current Law at node-1.

$$-100\angle 0^\circ \text{ m}+\frac{V-20\angle 0^\circ}{100}+\frac{V}{-j200}+\frac{V}{100+j100}=0 \quad \ldots\ldots (1)$$

$$-100\angle 0^\circ \text{ m}+\frac{V}{100}-\frac{20\angle 0^\circ}{100}+\frac{V}{-j200}+\frac{V}{100+j100}=0$$

$$V\left(\frac{1}{100}+\frac{1}{-j200}+\frac{1}{100+j100}\right)=0.1\angle 0^\circ+0.2\angle 0^\circ$$

$$V\left(0.01+j0.005+0.005-j0.005\right)=0.3\angle 0^\circ$$

$$V\left(0.015\angle 0^\circ\right)=0.3\angle 0^\circ$$

$$V=\frac{0.3\angle 0^\circ}{0.015\angle 0^\circ}$$

$$V=20\angle 0^\circ \text{ V}$$

Therefore, the value of the voltage V is $20\angle 0^\circ$ V.

Step 3 of 9

Calculate the value of the current through the inductor.

$$I_L = \frac{V}{100 + j100} \quad \text{...... (2)}$$

Substitute $20\angle 0°\ \text{V}$ for V.

$$\begin{aligned} I_L &= \frac{20\angle 0°}{100 + j100} \\ &= \frac{20\angle 0°}{141.42\angle 45°} \\ &= 0.141\angle -45°\ \text{A} \end{aligned}$$

Therefore, the value of the current passing through the inductor is $0.141\angle -45°\ \text{A}$.

Step 4 of 9

Calculate the value of the voltage across the inductor.

$$V_L = (100 + j100) I_L \quad \text{...... (3)}$$

Substitute $0.141\angle -45°\ \text{A}$ for I_L.

$$\begin{aligned} V_L &= (141.42\angle 45°)(0.141\angle -45°) \\ &= (141.42\angle 45°)(0.141\angle -45°) \\ &= 19.94\ \text{V} \end{aligned}$$

Therefore, the value of the voltage across the inductor is $19.94\ \text{V}$.

Step 5 of 9

Calculate the value of the reactive power in the inductor.

$$\text{Q}_L = \text{Im}\left(V_L I_L^*\right) \quad \text{...... (4)}$$

Substitute $19.94\ \text{V}$ for V_L and $0.141\angle -45°\ \text{A}$ for I_L.

$$\begin{aligned} \text{Q}_L &= \text{Im}\left(19.94 \times (0.141\angle -45°)^*\right) \\ &= \text{Im}(19.94 \times 0.141\angle 45°) \\ &= \text{Im}(2.81\angle 45°) \\ &= \text{Im}(2 + j2) \end{aligned}$$

$$\text{Q}_L = 2\ \text{VAR}$$

Therefore, the value of the reactive power in the inductor is $\boxed{2\ \text{VAR}}$.

Step 6 of 9

Calculate the value of the current through the capacitor.

$$I_C = \frac{V}{-j200} \text{ ...... (5)}$$

Substitute $20\angle 0° \text{ V}$ for V.

$$\begin{aligned} I_C &= \frac{20\angle 0°}{-j200} \\ &= \frac{20\angle 0°}{200\angle -90°} \\ &= 0.1\angle 90° \text{ A} \end{aligned}$$

Therefore, the value of the current passing through the capacitor is $0.1\angle 90° \text{ A}$.

Step 7 of 9

Calculate the value of the reactive power in the capacitor.

$$Q_C = \text{Im}\left(VI_C^*\right) \text{ ...... (6)}$$

Substitute $20\angle 0° \text{ V}$ for V_L and $0.1\angle 90° \text{ A}$ for I_L.

$$\begin{aligned} Q_C &= \text{Im}\left(20\angle 0° \times \left(0.1\angle 90°\right)^*\right) \\ &= \text{Im}\left(20 \times 0.1\angle -90°\right) \\ &= \text{Im}\left(2\angle -90°\right) \\ &= \text{Im}\left(0 - j2\right) \end{aligned}$$

$$Q_C = -2 \text{ VAR}$$

Therefore, the value of the reactive power in the capacitor is $\boxed{-2 \text{ VAR}}$.

Step 8 of 9

MATLAB calculations:

Use equations (1), (2), (3), (4), (5) and (6) in MATLAB to solve the problem.

Write the following instructions in MATLAB m-file and save the program.

```
syms V
V = solve(-0.1+((V-20)/100)+(V/-200i)+(V/(100+100i)))%NODAL EQUATION%
IL = V/(100+100i)%INDUCTOR CURRENT%
VL = IL*(100+100i)%INDUCTOR VOLTAGE%
QL = imag(VL*conj(IL))%INDUCTOR REACTIVE POWER%
IC = (V/-200i)%CAPCITOR CURRENT%
QC = imag(v*conj(IC))%CAPACITOR REACTIVE POWER%
```

Step 9 of 9

Run the program and view the results in the command window.

V = 20

IL = 1/10 - i/10

VL = 20

QL = 2

IC = i/10

QC = -2

Therefore, the values of the reactive power in the inductor and the capacitor are $\boxed{2\ \text{VAR}}$ and $\boxed{-2\ \text{VAR}}$ respectively.